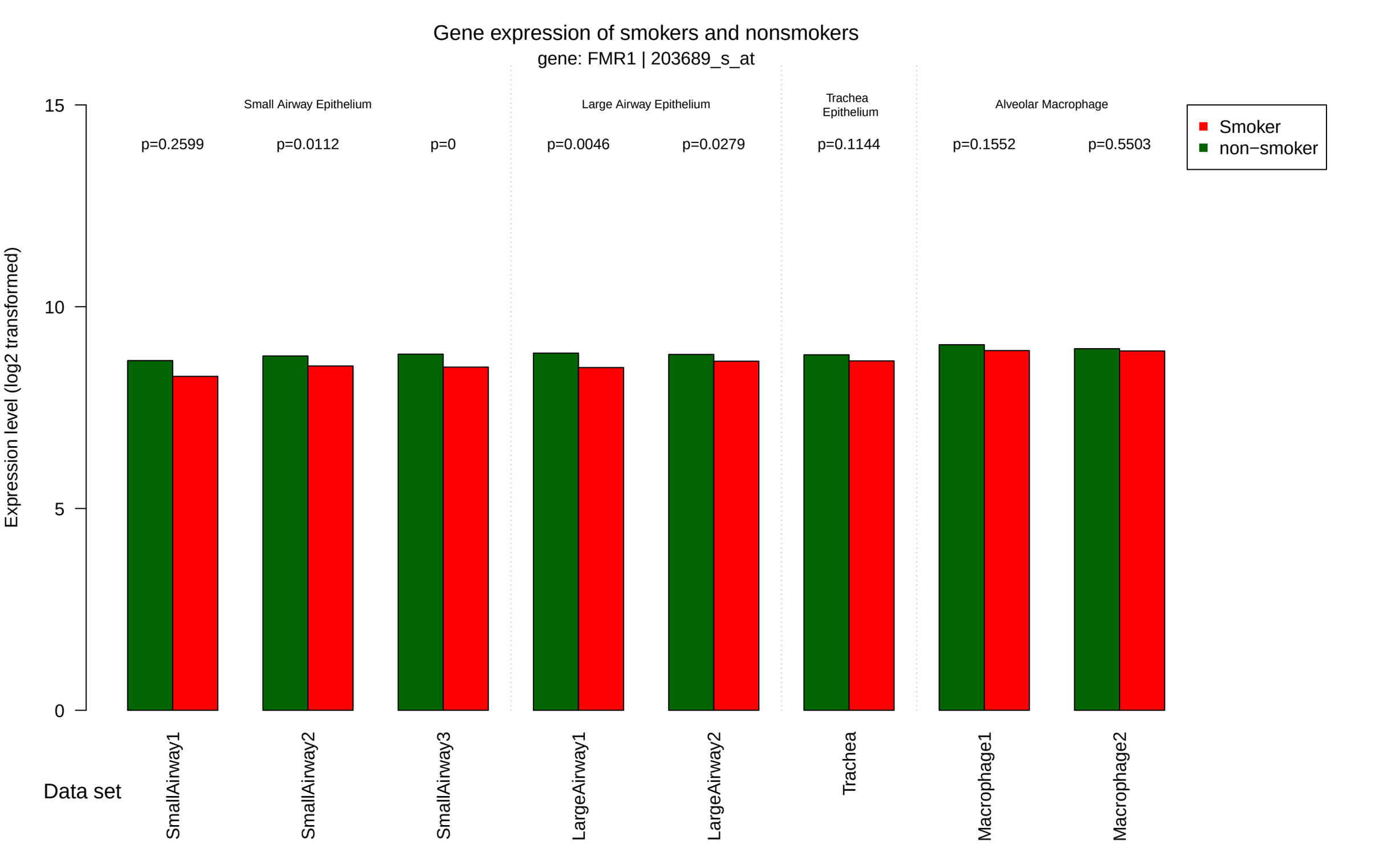
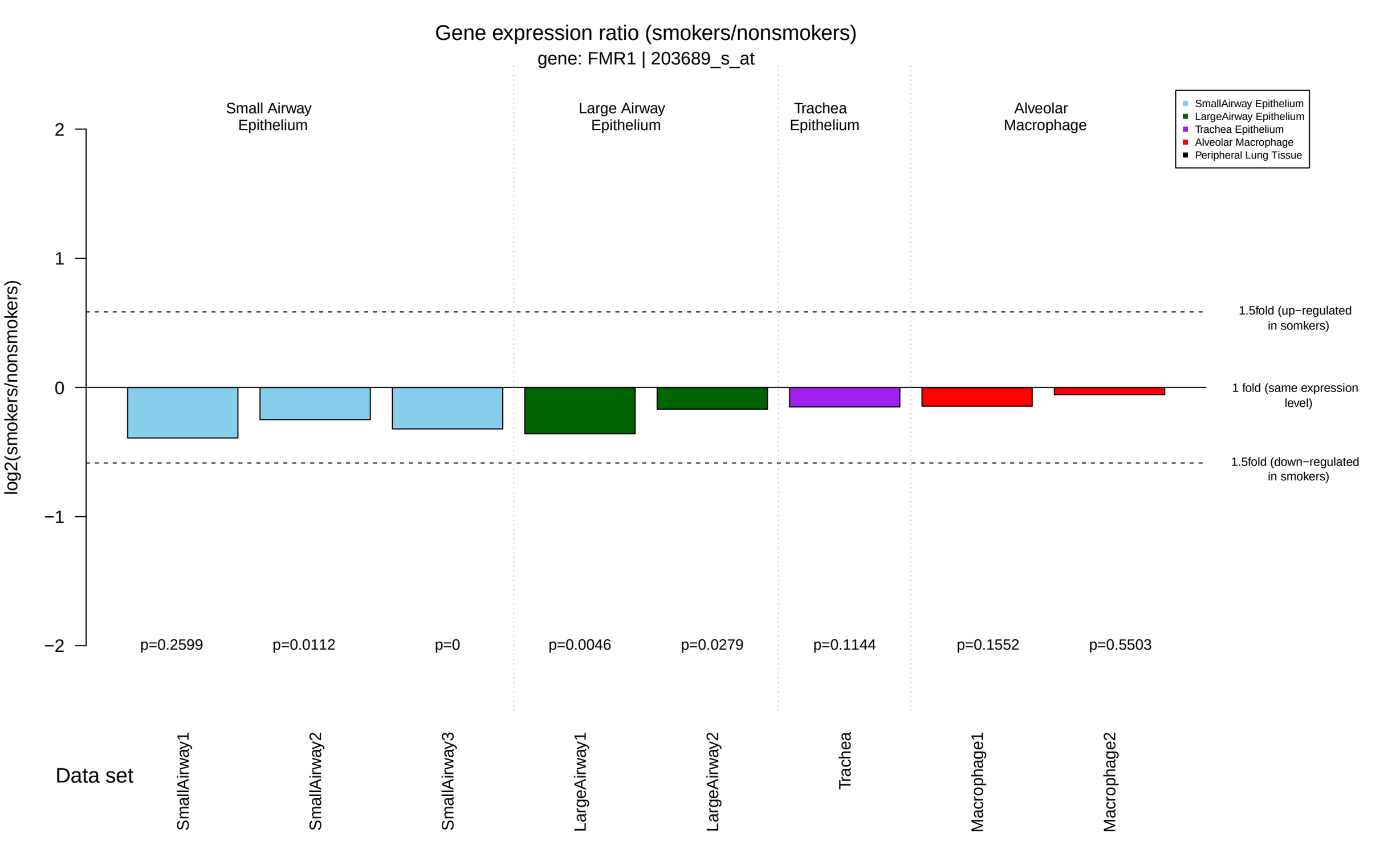
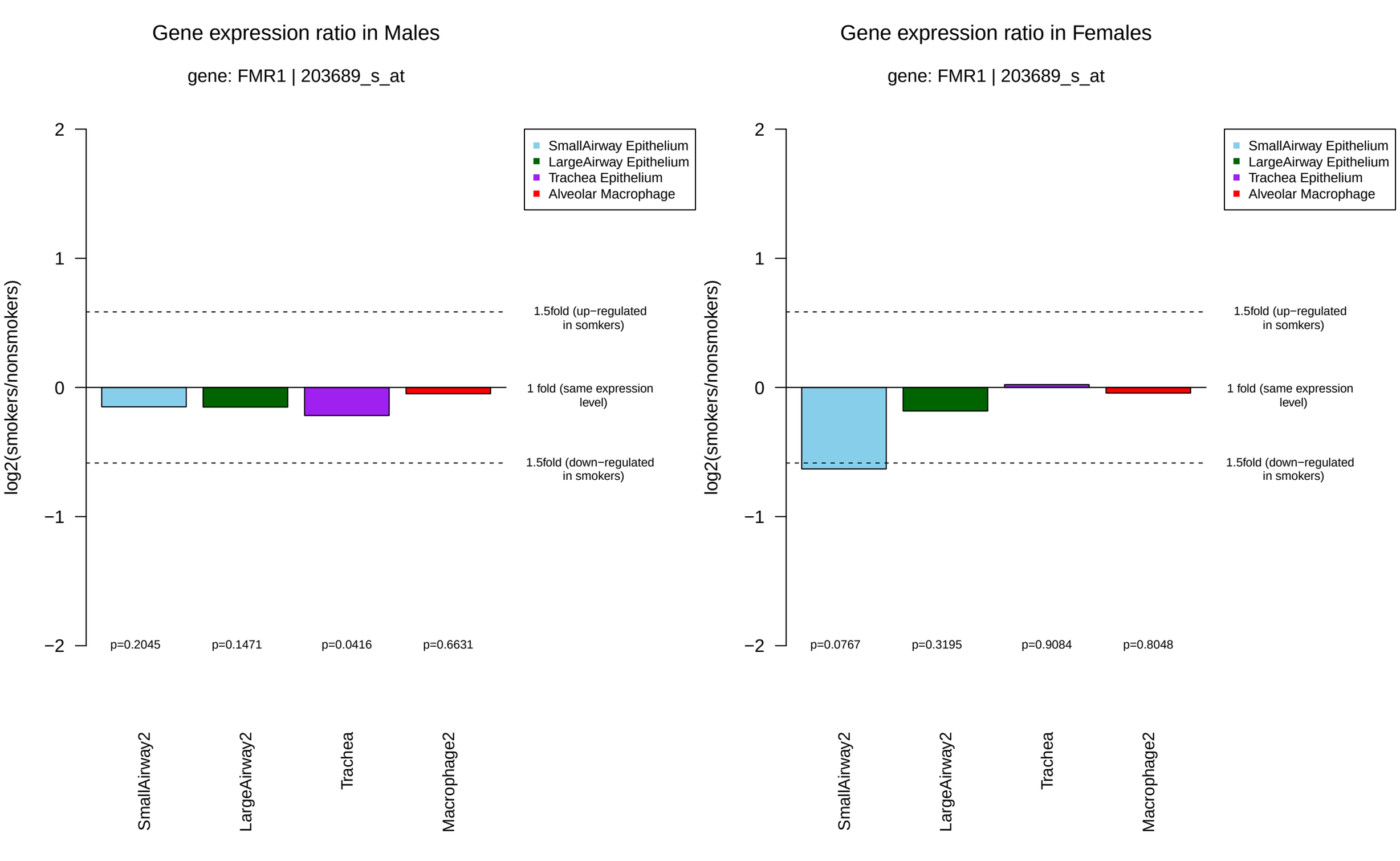
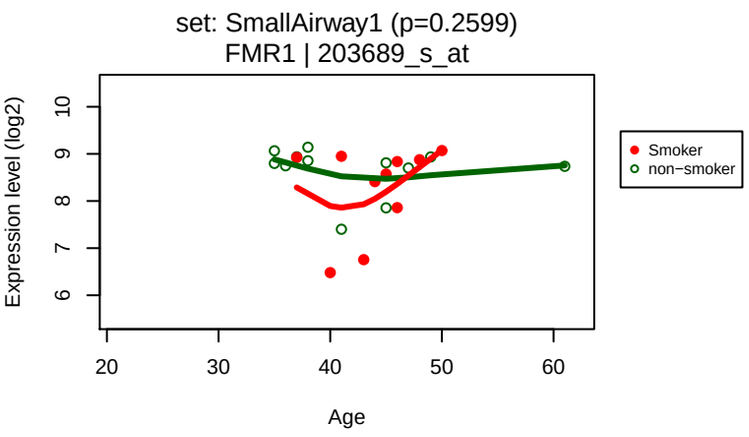
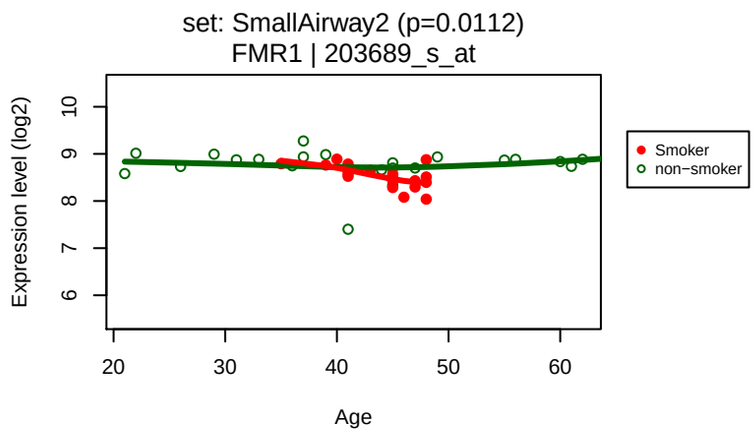
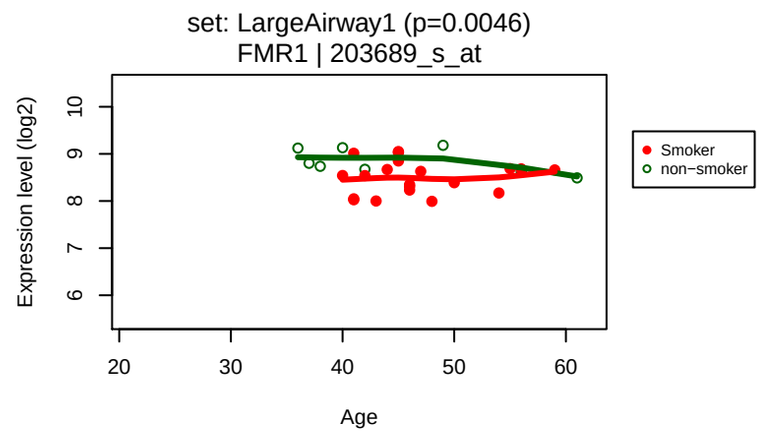

Gene expression of smokers and nonsmokers
gene: FMR1 | 203689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2599
p=0.0112
p=0
p=0.0046
p=0.0279
p=0.1144
p=0.1552
p=0.5503
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

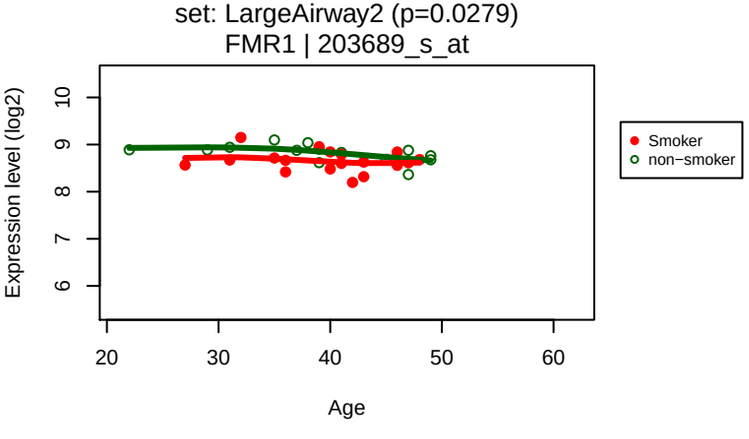

Gene expression ratio (smokers/nonsmokers)
gene: FMR1 | 203689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2599
p=0.0112
p=0
p=0.0046
p=0.0279
p=0.1144
p=0.1552
p=0.5503
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

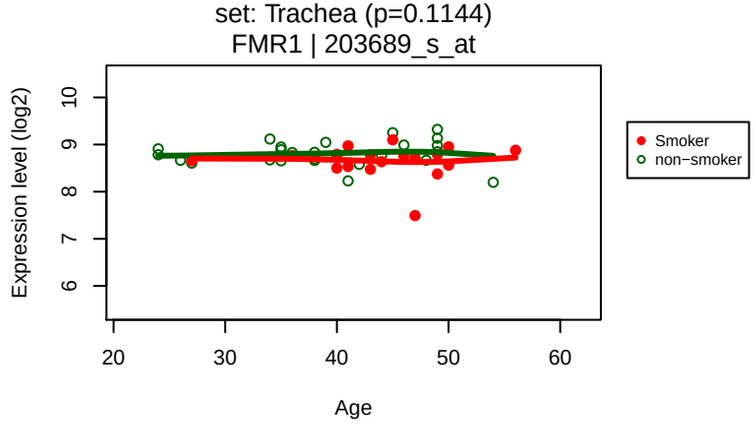

Gene expression ratio in Males
gene: FMR1 | 203689_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2045
p=0.1471
p=0.0416
p=0.6631
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FMR1 | 203689_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0767
p=0.3195
p=0.9084
p=0.8048
SmallAirway2
LargeAirway2
Trachea
Macrophage2

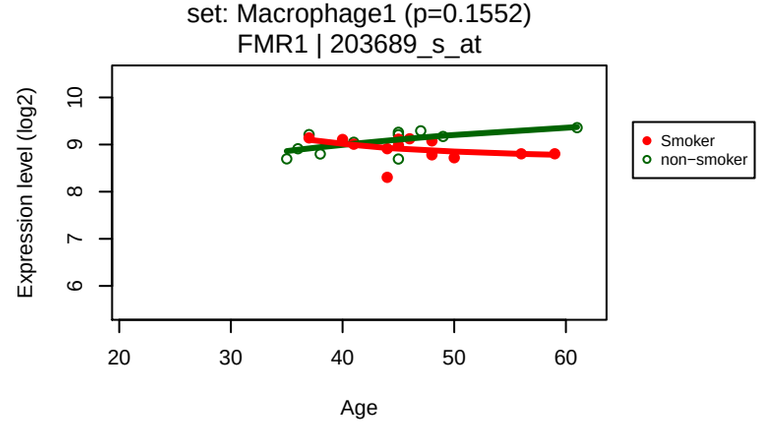

set: SmallAirway1 (p=0.2599)
FMR1 | 203689_s_at
Expression level (log2)
Age
Smoker
non−smoker

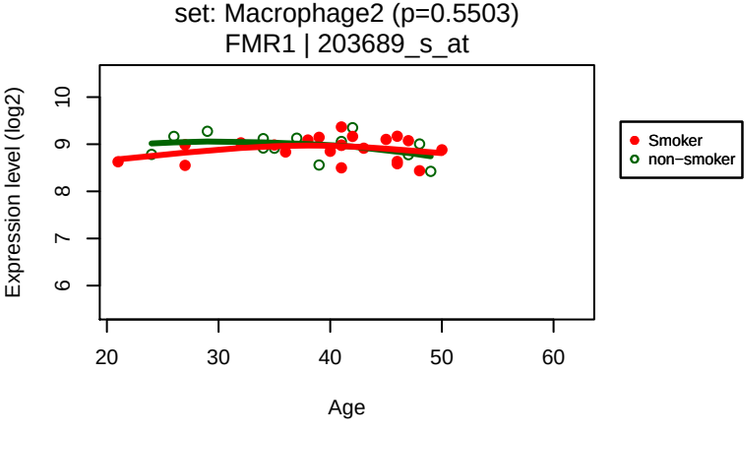

set: SmallAirway2 (p=0.0112)
FMR1 | 203689_s_at
Expression level (log2)
Age
Smoker
non−smoker

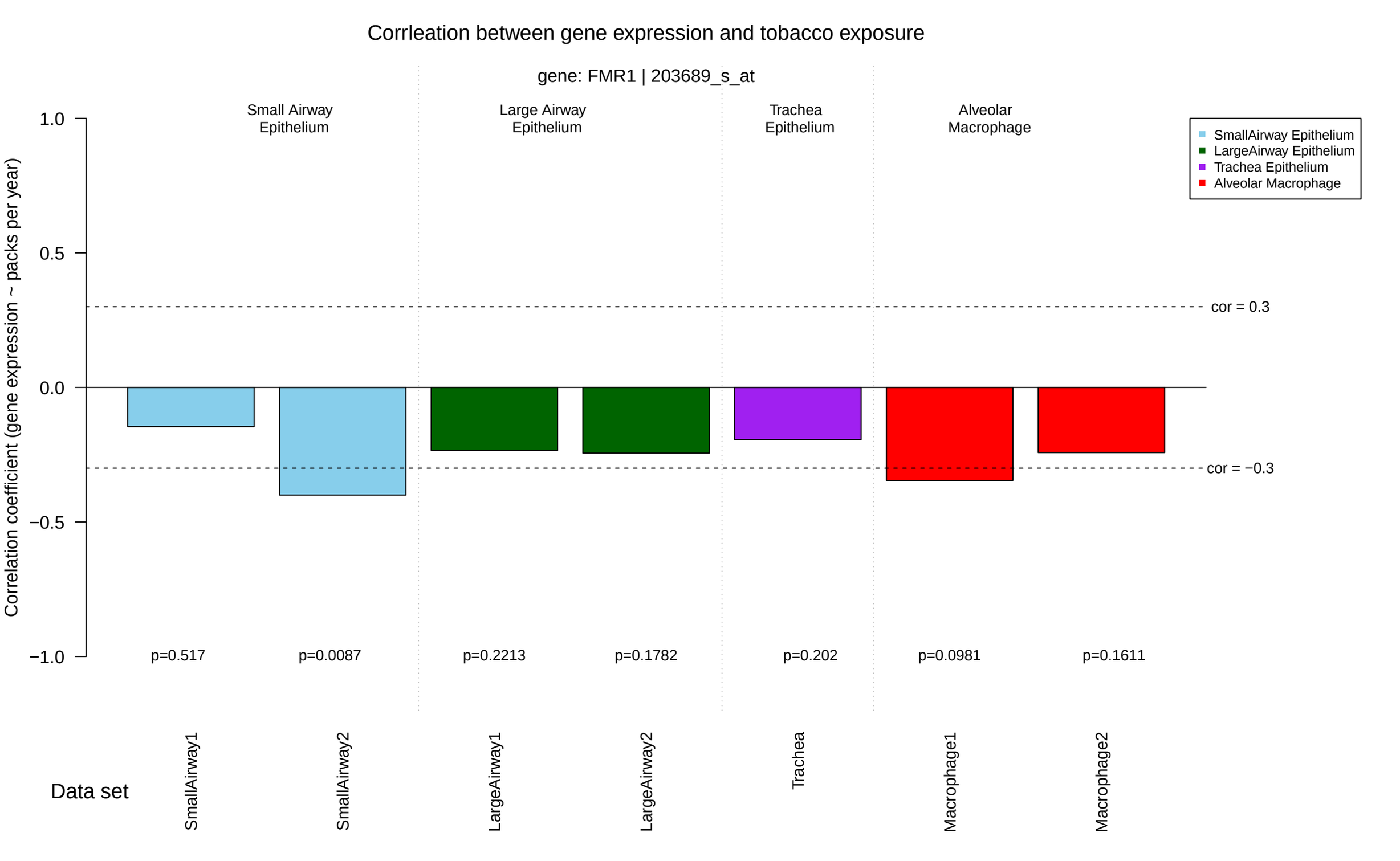

set: LargeAirway1 (p=0.0046)
FMR1 | 203689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0279)
FMR1 | 203689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1144)
FMR1 | 203689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1552)
FMR1 | 203689_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5503)
FMR1 | 203689_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FMR1 | 203689_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.517
p=0.0087
p=0.2213
p=0.1782
p=0.202
p=0.0981
p=0.1611
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2